COCOTE

Data Representation

- ***Data*** refers to the symbols that represent people, events, things, and ideas.
- Data can be a name, a number, the colors in a photograph, or the notes in a musical composition.
- ***Data Representation*** refers to the form in which data is stored, processed, and transmitted.
- Devices such as smartphones, iPods, and computers store data in digital formats that can be handled by electronic circuitry.

Number systems

- **Number Systems** are techniques used to represent numbers in the computer system architecture.
- Every value that you are saving or getting into/from computer has a defined number system.

- Computer architecture supports the following number systems.
  1. **Decimal Number System**
  2. **Binary Number System**
  3. **Octal Number System**
  4. **Hexadecimal Number System**

1. Decimal Number System

- **Decimal Number System** has only ten (10) digits from **0** to **9.**
- Every number represented with 0,1,2,3,4,5,6,7,8 and 9 is in this number system.
- The base of decimal number system is **10** because it has only **ten** digits.
- This number system is also referred to as **base-10** Number System.
- This is the number system that humans normally use.
- The prefix **deci** means tenth, hence the name.

2. Binary Number System

- Binary System has only two (2) digits: **0** and **1.**
- Every number (value) represented with 0 and 1 is this number system.
- The **base** of binary number system is **2** because it has only **two** digits.
- The 0s and 1s are referred to as ***bits*** (**bi**nary digi**ts**).
- This number system is also referred to as **base-2** Number System.
- For example, the number 2 is represented as 10 in binary number system.
- The prefix **bi** means two, hence the name.
- It is fundamental to representing and manipulating data in computers and other digital systems.
- Binary is used to store and process information at the lowest level in computing devices.

- A group of **eight** bits is called ***a byte***, and is usually abbreviated as uppercase **B.**
- A group of **4** bits is called a ***nibble..***

3. Octal Number System

- Octal Number System has only eight (8) digits from **0** to **7.**
- Every number (value) represented with 0,1,2,3,4,5,6 and 7 is in this number system.
- The base of the Octal number system is **8** because it has only **eight** digits**.**
- This number system is also referred to as **base-8** Number System.
- For Example, the number 10 is represented as 12 in Octal Number System.
- The word **octal** means eight, hence the name.

3. Hexadecimal Number System

- Hexadecimal Number System has sixteen (16) alphanumeric values from **0** to **9** and **A** to **F.**
- Every number represented with **0,1,2,3,4,5,6,7,8,9,A,B,C,D,E** and **F** is in this number system.
- The **base** of hexadecimal number system is **16** because it has sixteen alphanumeric values.
- Here **A** is **10**, **B** is **11**, **C** is **12**, **D** is **13**, **E** is **14**, and **F** is **15.**
- This number system is also referred to as **base-16** Number System.
- Hexadecimal is extensively used in programming, especially for representing memory addresses, color codes, and others.
- It is also used in various technical fields like networking, cryptography, and assembly language programming.

Table showing decimal numbers and their equivalents in Binary, Octal and Hexadecimal

Decimal	Binary	Octal	Hex
0	0	0	0
1	1	1	1
2	10	2	2
3	11	3	3
4	100	3	3
5	101	5	5
6	110	6	6
7	111	7	7
8	1000	10	8
9	1001	11	9
10	1010	12	A
11	1011	13	B
12	1100	14	C
13	1101	15	D
14	1110	16	E
15	1111	17	F
16	10000	20	10
20	10100	24	14
30	11110	36	1E
40	101000	50	28
50	110010	62	32
60	111100	74	3C
70	1000110	106	46
80	1010000	120	50
90	1011010	132	5A
100	11001000	144	64
1000	1111101000	1750	3E8
2989	101110101101	5655	BAD

Number System Conversion

- Number Systems can be converted from one system to another and vice versa.

CONVERTING FROM BASE-10 TO OTHER BASES

- To convert Number System from **Decimal Number System** to **Any Other Number System** is quite easy.

- The following two steps can be followed to do this conversion:
  1. Divide the Number (**Decimal Number**) by the base of the target base system (in which you want to convert the number to : Binary (2), Octal (8), Hexadecimal (16)).
  2. Write the remainder in step 1 as a Least Significant Bit (**LSB**) to Step last as a Most Significant Bit (**MSB**).

1010110010

MSB LSB

Example 1

Convert 8_{10} to Binary number system (base 2).

d/b = divider/base
q = quotient
r = remainder

d/b		LSB
2	8	r 0
2	q 4	0
2	2	0
2	1	1
	0	MSB

- Therefore $8_{10} = 1000_2$

Example 2

Convert $(45)_{10}$ to Binary Number System (base 2).

2	45	1
2	22	0
2	11	1
2	5	1
2	2	0
2	1	1
	0	

- Therefore, $\mathbf{45}_{10} = 101101_2$

Example 3

Convert 804_{10} to Octal Number System (Base 8)

8	**804**	4
8	**100**	4
8	**12**	4
8	**1**	1
	0	

- Therefore, $804_{10} = 1444_8$

Example 4

Convert 1600_{10} to base-16 Number system

16	**1600**	0
16	**100**	4
16	**6**	6
	0	

- Therefore, $1600_{10} = 640_{16}$

Example 5

Convert 30_{10} to base-16 (Hexadecimal)

16	30	14
16	1	1
	0	

- On remainder, we have 1 and 14, so, since 14 in base-16 is represented with **E** ,

therefore, $\mathbf{30_{10} = 1E_{16}}$ **.**

CONVERTING FROM OTHER BASES TO BASE-10

- To convert Number System from **Any Other Number System** to **Decimal Number System** is quite easy.
- The following steps have to be followed:

1. Identify the base of the given number: Determine the **base** of the original number.
2. Evaluate **the place value**: Starting from the **rightmost** digit of the given number, assign a place value to each digit based on its position. The rightmost digit represents the base raised to the power of 0 , the next digit raised to power of 1 and so on.
3. Multiply each digit by its corresponding place value: Multiply each digit of the given number by its respective place value.
4. Sum up the results: Add up the products obtained from the previous step to get the decimal equivalent of the original number.

Example

1. **Converting from base-2(decimal) to base-10 (decimal)**
   - $\mathbf{100100}_2 = (2^5 x \mathbf{1}) + (2^4 x \mathbf{0}) + (2^3 x \mathbf{0}) + (2^2 x \mathbf{1}) + (2^1 x \mathbf{0}) + (2^0 x \mathbf{0})$

     $= 32 + 0 + 0 + 4 + 0 + 0$

     $= 36_{10}$

2. Converting from base-8(octal) to base-10 (decimal)

- $\mathbf{2040}_8 = (8^3 \times \mathbf{2}) + (8^2 \times \mathbf{0}) + (8^1 \times \mathbf{4}) + (8^0 \times \mathbf{0})$

  $= 1024 + 0 + 32 + 0$

  $= 1056_{10}$

3. Converting from base-16(hexadecimal) to base-10 (decimal)

Convert $1E0D_{16}$ to base-10

$$1E0D_{16} = (16^3 \times 1) + (16^2 \times E) + (16^1 \times 0) + (16^0 \times D)$$

$$= (4096 \times 1) + (256 \times 14) + (16 \times 0) + (1 \times 13)$$

$$= 4096 + 3584 + 13$$

$$= 7693_{10}$$

Converting from base 2 to base 8 and vice versa.

- To convert from base-2 to base-8:
    1. Group the binary digits: Starting from the rightmost digit of the binary number, group the digits into **sets of three**. If there are not enough digits to form a complete group of three, **add leading zeros to the left**.
    2. Assign octal digits to each group: Replace each group of three binary digits with the corresponding octal digit.
    3. Combine the octal digits: Write down the octal digits obtained from the previous step, left to right. This represents the equivalent base-8 number.

e.g

- Convert 110111101_2 to base-8 (octal)
- Answer:

Group the binary digits from the rightmost into sets of 3

110 111 101

Replace each group of three binary digits with the corresponding octal digit

*110 -> 6, 111- >7, 101->**5***

Combine the octal digits

675

Therefore, $110111101_2 = 675_8$

Try converting 10101010_2 to base-8

Converting base-8 number system back to base-2

1. Identify the digits: Starting from the leftmost digit of the base-8 number, identify each digit individually.
2. Convert each digit to binary: Convert each octal digit to its equivalent binary representation.
3. Combine the binary digits: Write down the binary digits obtained from the previous step, left to right. This represents the equivalent base-2 number.

E.g

- Convert 543_8 to Binary Number System (base-2)

1. ***Identify the digits, i.e 5,4 and 3***
2. ***Convert each digit to binary***

   5 -> 101, 4 -> 100, 3 -> 011

3. ***Combine binary digits***

   101100011_2

Try: Convert 254_8 to base-2

Converting from base 2 to base 16 and vice versa.

Converting from base-2 to base-16:

1. Group the binary digits: Starting from the rightmost digit of the binary number, group the digits into sets of **four**. If there are not enough digits to form a complete group of four, add leading zeros to the left.
2. Assign hexadecimal digits to each group: Replace each group of four binary digits with the corresponding hexadecimal digit.
3. Combine the hexadecimal digits: Write down the hexadecimal digits obtained from the previous step, left to right.

e.g

- **Convert the binary number 10101110_2 to base-16 (hexadecimal)**

- Step 1: Group the binary digits: 1010 1110
- Step 2: Assign hexadecimal digits, 1010 ->10->**A**, 1110 -> 16 -> **E**
- Step 3: Combine the hexadecimal digits = **AE**

Therefore, $\mathbf{10101110}_2$ to base-16 is $\mathbf{AE}_{16}$

Try converting 0101111_2 to base-16

Converting from base-16 to base-2:

1. Identify the hexadecimal digits: Starting from the leftmost digit of the base-16 number, identify each digit individually.
2. Convert each hexadecimal digit to binary: Convert each hexadecimal digit to its equivalent binary representation.
3. Combine the binary digits: Write down the binary digits obtained from the previous step, left to right. This represents the equivalent base-2 number.

example

- Convert $2F_{16}$ to base-2

- **Step 1: Identify the digits: 2 and F**
- **Step 2: Convert each digital to binary, 2 -> 0010, F -> 1111;**
- **Step 3: Combine binary digits**

$$=00101111_2$$

TIP

- *To convert number system from binary to bases 8 and 16, you can first convert the binary digits to base-10 (decimal) and finally convert it further to base 8 or 16.*

- *Try to convert (using the tip)*

1. **10101110_2** to base-16
2. **110101010_2** to base-8

Converting to base-8 from base-16 and vice versa

Converting from base-16 to base-8

- **To convert from base-16 to base-8:**
  1. **First, convert each digit of base-16 number to base-2**
  2. **Then convert base-2 to base-8**

Example:

- Convert AC_{16} to base-8

- Since we have letters, we map them to their Hexadecimal equivalents
- **A-**>10, **C-**>12
- **Converting numbers to bits(binary digits)**
  A->10->**1010,** C->12->**1100**
  = 10101100

- **Convert the bits to base-8**
  010 101 100 (Group the bits into three sets)
- **Convert the bits to their Octal equivalents (or Decimal equivalents)**
  010 ->2, **101-**>5, **100-**>4
- **Group the digits**
  254

Therefore, AC_{16} = 254_8

Converting from base-8 to base-16

- **To convert from base-8 to base-16:**
  1. **First, convert each digit of base-8 number to base-2**
  2. **Then convert base-2 to base-16**

Example

- Convert **756_8** to Hexadecimal number system(base-16)

- **Identify and convert each digit to their binary equivalents.**
  7 -> **111**, 5 -> **101**, 6 -> **110**
- **Group the bits into set of four (nibble) starting from the right.**
  0001 1110 1110 (add preceding 0(s) if the set is not complete)
- **Convert the bits to their Hexadecimal equivalents**
  0001 -> **1**, 1110 -> **14 -> E**, 1110 -> **14 -> E**
- **Group the numbers**
  1EE
- **Therefore, 756_8 = $1EE_{16}$**

Representing texts

- Each computer has a set of characters that it uses.
- As a bare minimum, this set includes the **26 uppercase** letters, the **26 lowercase** letters, the digits **0** through **9**, and a set of special symbols, such as space, period, minus sign, comma, and carriage return.
- Each character has to be signed a number when being transferred to another computer.
- The mapping of characters onto integers is called a **character encoding**.
- It is essential that communicating computers use the same code or they will not be able to understand one another.
- For this reason, character encoding standards have been developed.

- Some of these standards include:

1. **ASCII** (American Standard Code for Information Interchange).
2. **EBCDIC** (Extended Binary Coded Decimal Interchange Code).
3. **Unicode**.

1. ASCII

- ASCII is a character encoding scheme that represents text in computers and other communication devices.
- It was developed in the 1960s by American Standards Association (ASA) and became the most widely used character encoding in the early days of computing.
- ASCII uses **7** bits to represent **128** different characters, including alphabets (both uppercase and lowercase).
- It was meant for **English alphabet only**, punctuation marks, and some special characters.
- The ASCII code for an uppercase **A** is **1000001**.

- Many of the ASCII control characters are intended for data transmission.
- For example, a message might consist of an **SOH** (Start of Header) character, a header, an **STX** (Start of Text) character, the text itself, an **ETX** (End of Text) character, and then an **EOT** (End of Transmission) character.
- In practice, however, the messages sent over telephone lines and networks are formatted quite differently, so the ASCII transmission control characters are not used much any more.

2. EBCDIC

- EBCDIC is a character encoding scheme primarily used on IBM mainframe computers and some legacy systems.
- Developed by IBM in the 1960s, EBCDIC uses 8 bits to represent characters, allowing for a larger character set compared to ASCII.
- EBCDIC supports a wide range of characters, including alphabets (uppercase and lowercase), numbers, symbols, and control characters.
- EBCDIC encoding differs significantly from ASCII, making it incompatible with ASCII-based systems without conversion.

3. UNICODE

- Unicode is a universal character encoding standard that aims to represent all characters from all writing systems used worldwide.
- It was introduced in 1991 and maintained by Unicode Consortium.
- It provides a consistent way to encode and exchange text data across different platforms, languages, and devices.
- Unicode uses a variable-length encoding scheme, with different encoding formats like UTF-8, UTF-16, and UTF-32.
- UTF-8 is the most commonly used encoding format, capable of representing the entire Unicode character set using 8-bit code units.
- Unicode supports a vast range of characters, including characters from various languages, symbols, emojis, mathematical symbols, and special characters.
- It enables multilingual text representation and allows for seamless communication and interoperability between different systems and languages